APIs, Data Collection & NLP

REST APIs, HTTP Requests, File Handling, Python Frameworks, NLTK, and ChatterBot

by

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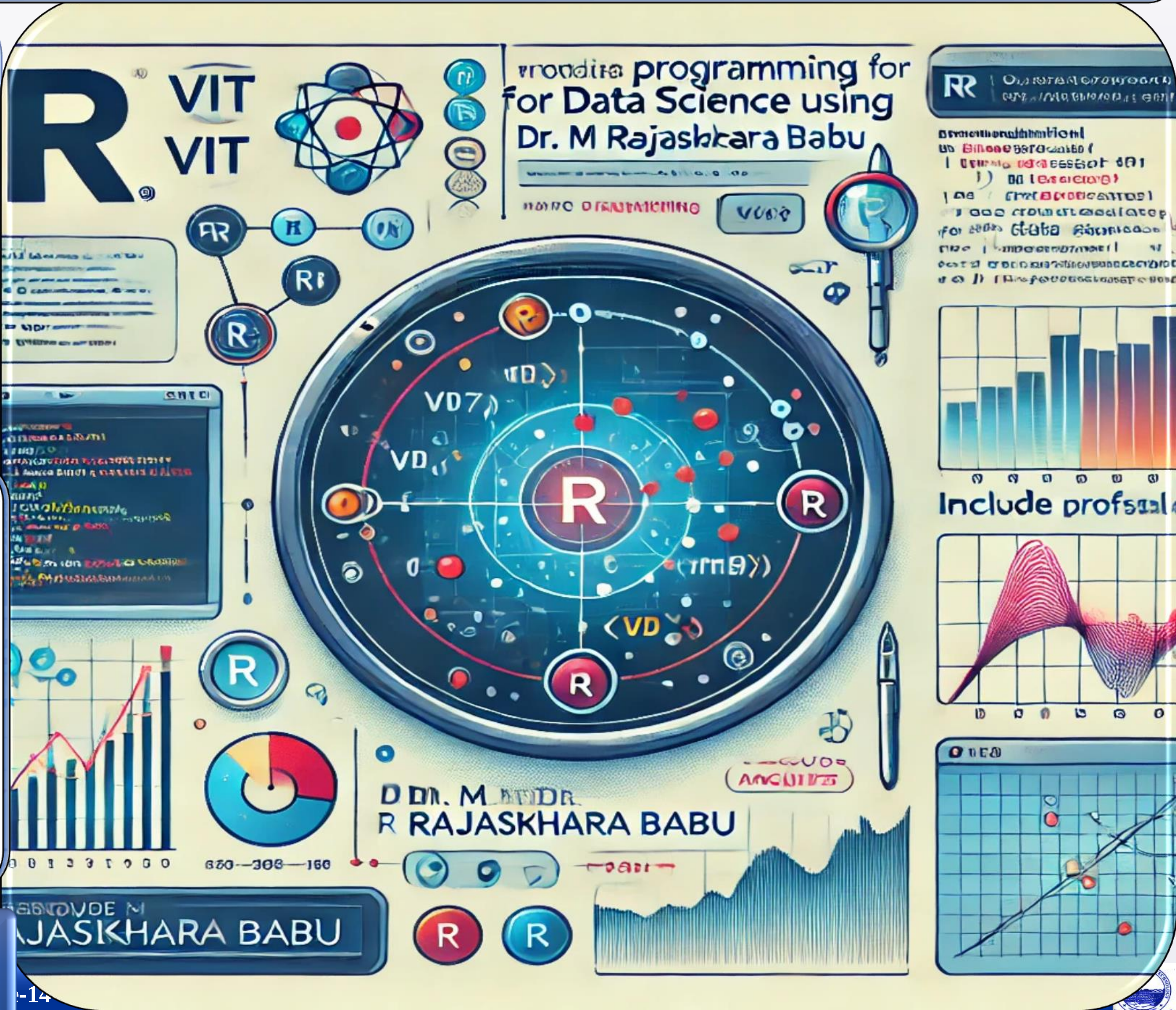


Objectives & Teaching Learning Material

To explain the concepts of data collection, APIs, REST services, and HTTP requests for acquiring data from various sources

To demonstrate file handling techniques and the use of Python libraries for processing structured and unstructured data

To introduce Natural Language Processing concepts using NLTK and ChatterBot for text analysis and chatbot development

Teaching + Learning **Material:** LCD, White board Marker, Presentation slides

Learning outcomes

Upon successful completion of this lecture, students will be able to:

- Collect and retrieve data from APIs, REST services, and files using appropriate Python tools and techniques

- Process structured and unstructured data using file handling methods and Python libraries such as Requests, Pandas, and NLTK

- Develop basic NLP applications involving text preprocessing, language analysis, and chatbot development using NLTK and ChatterBot

Session Plan

Time in minutes	Topic	Learning Aid & Methodology	Faculty Approach	Student Activity	Skill and Competency Developed
05	Re-Cap	Quiz Presentation	Questions Organizes	Answers Identifies	Knowledge Understanding
10	Data Collection, APIs, and REST Services	Presentation	Presentation	Listens	Remembering
10	HTTP Requests and File Handling	Explains	Explains	Notes	Understanding
15	Python Frameworks and Natural Language Processing	Case Study	Organizes	Observes	Applying
10	NLTK, ChatterBot, and Real-World Applications	Discussion	Facilitates	Answers	Analyzing

What is Data Collection?

- Data collection is the process of gathering information from various sources for analysis, visualization, and decision-making.
- It is the foundation of every Data Science pipeline.

Sources of Data

- Files (CSV, JSON, Excel)
- Databases & APIs
- Websites & Social Media
- Sensors and IoT Devices
- Enterprise Applications

Why It Matters

- Provides raw data for analysis
- Supports machine learning models
- Enables predictive analytics
- Helps derive actionable insights

Data Collection Workflow

Every data science project follows a structured pipeline from raw source to actionable insight.

Data Source | **Collection** | **Storage** | **Processing** | **Analysis**

Each stage transforms raw information into structured, usable knowledge — enabling reliable analysis and decision-making downstream.

What is an API?

API stands for Application Programming Interface — a bridge that allows different software applications to communicate and exchange data.

Think of an API like a waiter: it carries requests from the customer (client) to the kitchen (server) and returns the response.

Weather API
Real-time weather data

Google Maps API
Location & routing

OpenAI API
AI-powered responses

Payment APIs
Secure transactions

Why Use APIs?

Real-Time Data

Access live, up-to-date information from external services instantly.

Automate Collection

Eliminate manual data gathering with programmatic API calls.

Integrate Systems

Connect different applications and cloud-based services seamlessly.

External Systems

Retrieve data from Twitter, News, Maps, and other external platforms.

REST APIs

- **REST (Representational State Transfer) is the most widely used API architecture,**
- **enabling communication between applications over the web using HTTP protocols.**

Key Characteristics

- Client-Server Communication
- Stateless Communication
- Standard HTTP Methods
- Supports JSON and XML
- Scalable and Lightweight

Common Endpoints

- `/api/students`
- `/api/courses`
- `/api/weather`
- `/api/products`

The client sends an HTTP Request; the server returns JSON or XML data.

HTTP Requests

HTTP (HyperText Transfer Protocol) is the communication protocol used between clients and servers on the internet. Python's `requests` library makes it simple.

Method	Purpose
GET	Retrieve data from a remote server
POST	Send / create new data on the server
PUT	Update existing data on the server
DELETE	Remove data from the server

```
import requests
response = requests.get("https://api.example.com/data")
print(response.json()) # GET example


data = {"name": "John"}
requests.post(url, json=data) # POST example
```

API Response Formats

APIs return data in structured formats. JSON is the most commonly used format in Data Science applications, while XML is also supported.

JSON Format

```
{
  "city": "Tirupati",
  "temperature": 32
}
```

XML Format

```
<city>Tirupati</city>
<temperature>32</temperature>
```

JSON is preferred for its simplicity, readability, and native compatibility with Python dictionaries.

File Handling in Python

- Data Science projects constantly require reading datasets, writing processed data, saving results, and generating reports.
- Python's built-in file handling makes this straightforward.

File Modes

`r` Read | `w` Write | `a` Append | `r+` Read & Write

Best Practice: `with` Statement

Automatically closes the file, ensures better resource management, and produces cleaner code.

```
with open("data.txt", "r") as
file:
 content = file.read()
 print(content)
```

Reading & Writing CSV and JSON Files

CSV and JSON are the two most common file formats in Data Science workflows. Python handles both with minimal code.

CSV with Pandas

```
import pandas as pd

# Read
df =
pd.read_csv("students.csv")
print(df.head())

# Write
df.to_csv("output.csv",
          index=False)
```

JSON with json module

```
import json

# Read
with open("data.json") as f:
 data = json.load(f)

# Write
with open("output.json","w") as
f:
 json.dump(data, f)
```

Frameworks & Libraries

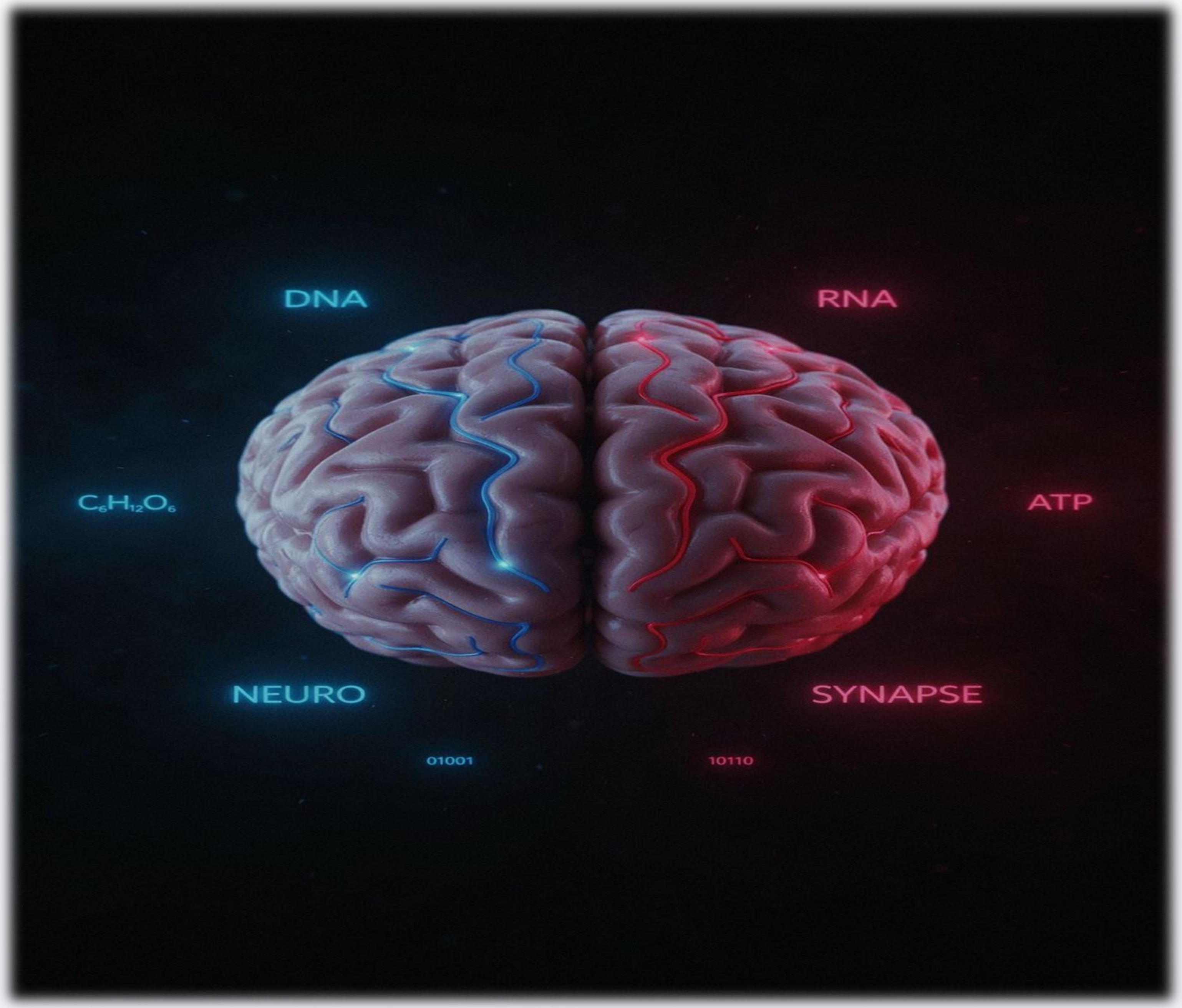
A library is a collection of pre-written code that helps programmers accomplish tasks more efficiently. Python's ecosystem is the industry standard for Data Science.

Library	Purpose
NumPy	Numerical Computing
Pandas	Data Analysis
Matplotlib / Seaborn	Data & Statistical Visualization
Requests	API Communication
NLTK	Natural Language Processing
Scikit-Learn	Machine Learning

Faster Development	**Reusable Code**
Industry Standard	**Reduced Dev Time**

Natural Language Processing (NLP)

NLP is a branch of Artificial Intelligence that enables computers to understand and process human language — bridging the gap between human communication and machine understanding.



Chatbots
Conversational AI assistants

Sentiment Analysis
Understanding opinion in text

Translation
Automated language translation

Speech Recognition
Converting spoken words to text

NLTK — Natural Language Toolkit

NLTK is a Python library designed for text processing, language analysis, linguistic research, and NLP. Install with `pip install nltk`.

1

Tokenization

Breaking text into smaller units (words or sentences).

```
from nltk.tokenize import word_tokenize
tokens = word_tokenize("Data Science is amazing.")
# ['Data','Science','is','amazing','.']
```

2

Stop Word Removal

Removing common words (*is, the, a, an, of*) that contribute little meaning.

```
from nltk.corpus import stopwords
stop_words = set(stopwords.words('english'))
```

Stemming & Lemmatization

Both techniques reduce words to a base form, but with different approaches and levels of linguistic accuracy.

Stemming

Reduces words to their root form (may not be a real word).

playing → play | running →

```
from nltk.stem import
PorterStemmer
stemmer = PorterStemmer()
print(stemmer.stem("running"))
# run
```

Lemmatization

Converts words into meaningful, dictionary base forms.

```
from nltk.stem import
WordNetLemmatizer
lemmatizer =
WordNetLemmatizer()
print(lemmatizer.lemmatize("
running"))
```

ChatterBot

ChatterBot is a Python library used for building simple conversational chatbots. Install with `pip install chatterbot`.

Conversational AI

Auto Response Generation

Trainable Models

Easy Integration

Building & Training a ChatterBot

Creating and training a ChatterBot requires just a few lines of Python. The `ListTrainer` teaches the bot with conversation pairs.

Create & Train

```
from chatterbot import ChatBot
from chatterbot.trainers import
ListTrainer

bot = ChatBot("Assistant")
trainer = ListTrainer(bot)
trainer.train([
    "Hello",
    "Hi, how can I help you?"
])
```

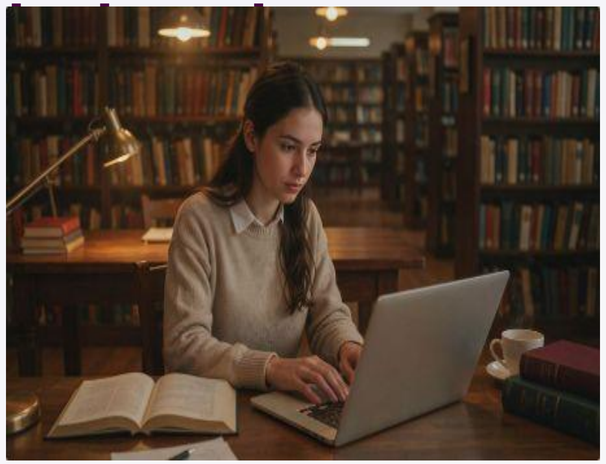
Get a Response

```
response =
bot.get_response("Hello")
print(response)
# Hi, how can I help you?
```

The bot learns from training data and generates contextually appropriate responses to user inputs.

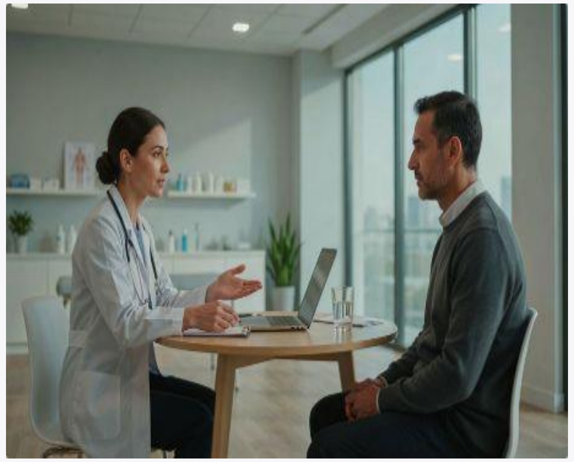
ChatBot Applications

Conversational chatbots are transforming service delivery across multiple



Education

Student assistance and learning support systems on smart campuses.

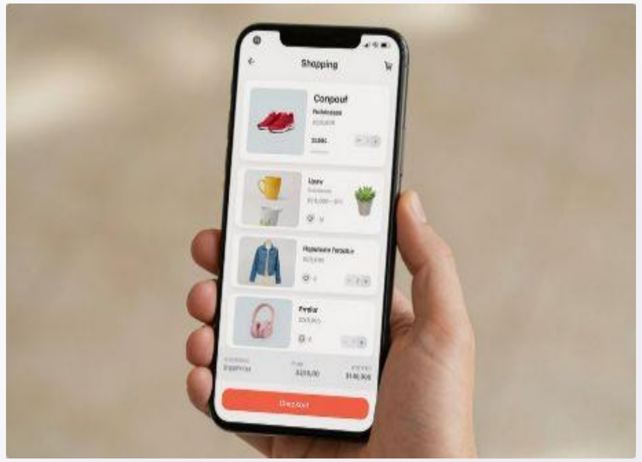


Healthcare

Patient guidance, appointment scheduling, and symptom checking.

Banking

24/7 customer support and account query resolution.



E-Commerce

Personalized product recommendations and order tracking.

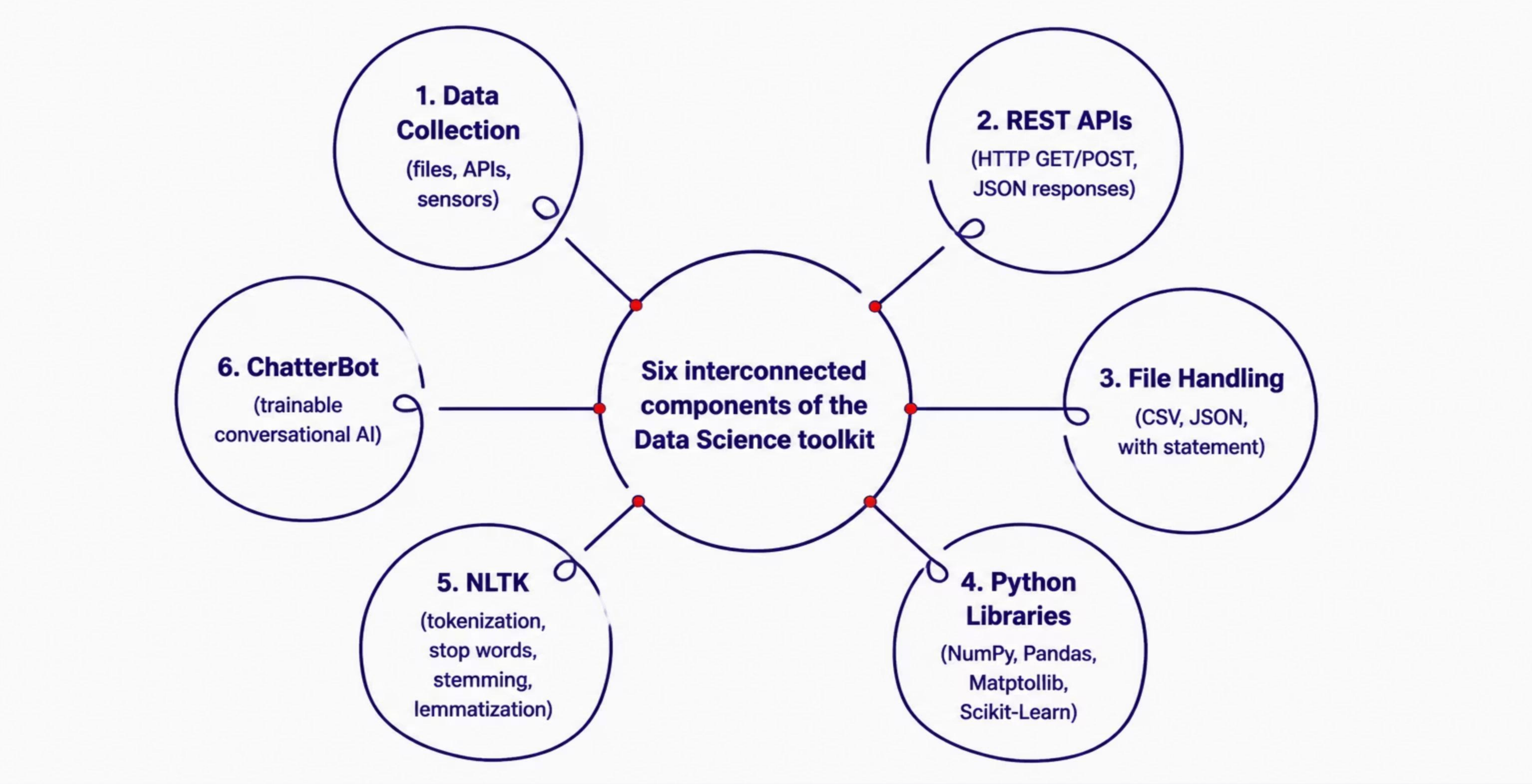
NLTK vs. ChatterBot

Both libraries serve the NLP ecosystem but with distinct purposes — NLTK for text processing and ChatterBot for conversational AI development.

Feature	NLTK	ChatterBot
Primary Purpose	NLP Processing	Chatbot Development
Text Processing	✓ Yes	Limited
Tokenization	✓ Yes	✕ No
Stemming	✓ Yes	✕ No
Chatbot Support	Indirect	✓ Yes
Training Capability	✓ Yes	✓ Yes

The Full Data Science Toolkit

This lecture covered the complete pipeline — from collecting data via APIs and files, to processing language with NLTK and building chatbots with ChatterBot.



Key Takeaways

1 APIs enable real-time data access

REST APIs use HTTP methods (GET, POST, PUT, DELETE) and return JSON or XML.

2 File handling is essential

Use the `with` statement for safe reading/writing of CSV and JSON files.

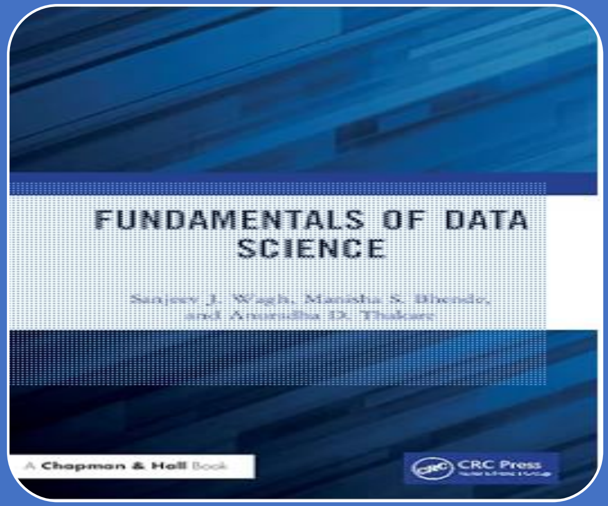
3 NLTK powers text analysis

Tokenization, stop word removal, stemming, and lemmatization are core NLP tasks.

4 ChatterBot enables conversational AI

Trainable chatbots can be deployed across education, healthcare, banking, and e-commerce.

Reference



T1. Sanjeev Wagh, Manisha Bhende, Anuradha Thakare, ‘Fundamentals of Data Science, CRC Press, 1 st Edition, 2022.

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